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The acceptance of COVID-19 vaccines in Rwanda: a cross-sectional study

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Abstract

The first reported case of Coronavirus Disease 2019 (COVID-19) in Rwanda occurred on March 14 2020. By the end of July 2024, a total of 133,518 individuals had tested positive for the infection, resulting in 1,468 deaths and 132,039 had fully recovered. The success of COVID-19 elimination in Rwanda hinges on the public's level of acceptance of the COVID-19 vaccination. Although COVID-19 is no longer a pandemic anymore, the World Health Organisation recommends countries vaccinate their populations to protect them from COVID-19 and its variants. Globally, COVID-19 has affected 704,753,890 people, caused 7,010,681 deaths and 675,619,811 have recovered. This study aimed to assess the acceptability of COVID-19 vaccines among adults aged 18 years and above in Rwanda. A cross-sectional study was conducted from January to March 2022 to determine the associations between COVID-19 vaccine acceptance (VA) with respondents' characteristics, using logistic regression analysis. This study enrolled 2,126 respondents with a mean age of 31 years, the majority of whom were females (82.2%), 51.4% had completed primary education, and 78.7% were married. Most respondents recognized the importance of COVID-19 vaccination for both personal health and community well-being. The study found a high rate of COVID-19 vaccine acceptance, with 91.6% of respondents expressing VA and an overall VA rate of 98.2%. Having a relationship with the child(ren) was the only characteristic associated with COVID-19 vaccine acceptance (p ; 3.2×10^{-3} , OR; 2.9, 95% C.I; 1.4–5.9). In conclusion, the study found a high rate of COVID-19 vaccine acceptance among adults in Rwanda, with COVID-19 associated with having a relationship with the child(ren). The study recommends the need for mass educational campaigns and awareness-raising efforts to understand of COVID-19 vaccines.

Keywords COVID-19, Vaccination, Acceptability, Vaccine acceptance, Rwanda

Introduction

The Coronavirus Disease of 2019 (COVID-19) pandemic, caused by the Severe Acute Respiratory Syndrome Coronavirus represents a serious threat to global health, the economy and security [1, 2]. The COVID-19 pandemic has inflicted severe damage on both the world's most

developed countries and the low- and middle-income countries of Africa [2].

Since the onset of the COVID-19 pandemic in December 2019, approximately 704,753,890 people have been infected globally by the end of July 2024 according to Worldometer.info [3], with 142,229 deaths reported across 213 countries [2]. As of July 2024, approximately 675,619,811 people had fully recovered with around seven million deaths globally [4].

To mitigate the adverse health effects of COVID-19, countries have implemented various preventive

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measures, including the use of facemasks, community lockdowns, and stay-at-home orders [5], and the promotion of hygiene practices, to reduce transmission rates. Although these measures were effective, the reappearance of COVID-19, often with different variants, was reported in many areas after the resumption of normal activities [6, 7], highlighting the urgent need for a long-term preventive strategy.

Such a long-term strategy can only be achieved through vaccination, which is considered one of the most significant success stories in modern medicine. With the approval of many COVID-19 vaccines, recent efforts have focused on mass vaccination as a reliable and cost-effective means to halt the outbreak [8]. However, despite vaccines being offered free to the public, vaccine hesitancy has emerged as a major challenge.

The concept of vaccine hesitancy, which refers to the extent to which individuals delay or refuse vaccination [9–11], plays a crucial role in influencing the rate of vaccine acceptance and distribution. Evidence indicates that vaccine hesitancy can be attributed to the 5 C scale: confidence, complacency, constraints, calculation, and collective responsibility. The 5 C scale examines various psychological concepts, such as attitude (confidence), perceived personal health status and invulnerability (complacency), self-control (constraints), preference for deliberation (calculation), and communal orientation (collective responsibility), among others [9].

Vaccine confidence therefore is considered one of the primary factors influencing vaccine acceptability and uptake [12, 13]. It refers to an individual's level of trust in the vaccine's reliability and safety, as well as their trust in the healthcare professionals and policymakers [10, 11, 14]. The issue of vaccine safety has been addressed through collaborative efforts involving medical professionals, legislators, community activists, and government institutions. Widespread vaccination campaigns have been enhanced and achieved through efficient communication channels that rely on evidence-based studies [15].

Furthermore, the term “vaccine convenience” refers to the ease with which individuals can access and obtain vaccinations, considering factors such as vaccine portability, affordability, and distribution [11, 14]. Conversely, when the perceived risk of immunisation infections is minimal and individuals do not view vaccination as a crucial preventive measure, this is termed “vaccine complacency” [16].

In addition to the 5 C scale, context-specific factors contribute to the COVID-19 vaccine hesitancy in Africa. These factors include global COVID-19 vaccine inequality [17, 18], a lack of vaccine production facilities, insufficient awareness, insecurity, endemic corruption, mistrust in certain political leaders, the spread of unverified anti-vaccination rumours and misinformation [5], and

political instability. These challenges result in an overall mean COVID-19 vaccine acceptance rate of 58%, with vaccine hesitancy persisting across the African continent [13].

The Rwanda Ministry of Health (MoH) reported the first COVID-19 case in March 2020. Following this initial detection, the Rwandan government implemented substantial management measures, including a nationwide lockdown and mandatory use of facemasks [19, 20]. By mitigating infections threatening border protection, Rwanda has reinforced its healthcare system as a response to infectious disease outbreaks [21].

Learning from the devastating effects of the Ebola virus, Rwanda took proactive precautions to minimise the impact of potential equivalent occurrences, such as the COVID-19 pandemic. Researchers from the World Health Organisation (WHO) stationed in Rwanda observed how the country applied its knowledge to control both the COVID-19 and Ebola epidemics [22].

After the first case of COVID-19 was reported in Rwanda, 133,194 individuals tested positive and 1.1% (1,468 people) succumbed to the disease by September 2023 [4]. Rwanda is among the few countries in Africa that quickly adopted and implemented early mitigation measures and undertook progressive capacity-building activities for frontline healthcare workers to fight COVID-19 [23, 24]. Following the capacity building of frontline healthcare workers, Rwanda purchased 437 new refrigerators, refrigerated vehicles, passive containers for transportation, and five ultra-low temperature freezers to ensure readiness for the Pfizer vaccine, which needed to be stored at -70°C . This enhanced readiness allowed the country to store over five million vaccines before receiving the first batch of vaccines in March 2021.

The Rwandan MoH reported that 77.9% of its population had received all COVID-19 vaccinations as of September 2022 [25]. Despite this high vaccination rate, we are unaware of any studies specifically examining whether Rwandans are more or less likely to vaccinate themselves against COVID-19. However, studies have been conducted to assess the knowledge and attitudes of various groups in Rwanda towards the COVID-19 pandemic [26], as well as the impact of the pandemic on malaria services in three Districts in Rwanda [27]. To plan for future immunisation campaigns against COVID-19 and other vaccine-preventable infections, it is crucial to assess vaccine acceptance, as it plays a vital role in determining whether individuals will choose to get vaccinated. In this study, we investigated the acceptability of novel COVID-19 immunisation among adults in the Rwandan population.

Methodology

Study area, population, and sample size

Rwanda is a landlocked country in east-central Africa, renowned for its stunning scenery and referred to as the “Land of a Thousand Hills.” The capital city, Kigali, is situated in the central part of the country [28]. With a population of 14.15 million people and a growth rate of 2.31%, Rwanda has one of the highest population densities in sub-Saharan Africa. The Rwandan MoH has collaborated with local and international non-governmental organisations (NGOs) to identify factors that will increase immunisation adoption and achieve sector development goals, including reducing deaths among children under five by two-thirds.

Sample size

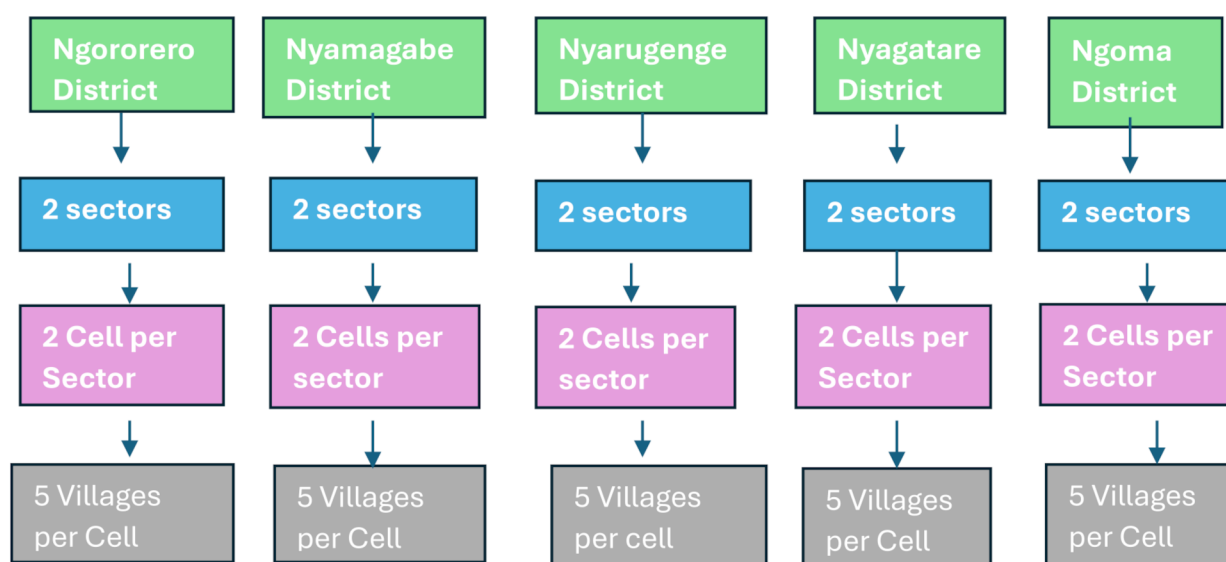
The sample size was calculated using the CDC Epi Info 7.2.5.0 (Centre for Disease Control, Georgia, USA) StatCalc. In the absence of similar studies in Rwanda, a minimum sample size of 154 per cluster/district was calculated based on the WHO immunisation coverage cluster survey [29]. The calculation considered the following characteristics: an estimated district population size of 362,806 in 2022 [30], an estimated proportion of acceptance of COVID-19 vaccines of 50.0%, a design effect of 2.0, an accepted error margin of 5%, and five clusters/

districts. To account for possible non-response, the sample size was increased by 10%, resulting in a total of 170 respondents per cluster/district.

Study design

The study utilised cross-sectional data from five randomly selected districts in different Rwandan Provinces, including Ngororero, Nyamagabe, Nyarugenge, Nyagatare, and Ngoma district. As shown in Fig. 1, each district is divided into sectors, which are further subdivided into Cells (or “Akagali”) and Villages.

Local governance in Rwanda is structured with municipalities overseeing district affairs, sector and cell executives managing administrative levels, and village leaders known as “Mudugudu,” maintaining governance at the village level. We utilised stratified cluster analysis. Ten sectors, comprising two from each district, were selected from five districts. Additionally, five villages were selected from each of the four cells, resulting in a total of 50 Rwandan Villages serving as our study sites. Using a random sample generator, we collected the household registration codes for the five districts/clusters, allowing us to complete the sampling procedure in three stages for the requisite number of respondents (Ngoma, Ngororero, Nyagatare, Nyamagabe, and Nyarugenge). Following the enumeration and calculation of



Legend

- 5/30 districts selected by RG
- 10/68 Sectors selected by RG
- 20/382 Cells selected by RG
- 50/2412 villages selected by RG

Fig. 1 Schematic illustration of the study settings using a multistage sampling method

Table 1 Questions regarding vaccine acceptance and coding scheme

Index	Category	Question	Answer	Coding
Q30	General vaccine	Vaccines are important for my health.	- Strongly Disagree, - Disagree, - Undecided/Neutral, - Agree - Strongly Agree	- Agree: 'Agree' and 'Strongly Agree' - Disagree: otherwise
Q36	General vaccine	Getting vaccinated is a good way to protect myself from disease.	- Strongly Disagree, - Disagree, - Undecided/Neutral, - Agree, - Strongly Agree	- Agree: 'Agree' and 'Strongly Agree' - Disagree: otherwise
Q44	General vaccine	Getting myself vaccinated with a COVID-19 vaccine is important for the health of others in my community.	- Strongly Disagree, - Disagree, - Undecided/Neutral, - Agree, - Strongly Agree	- Agree: 'Agree' and 'Strongly Agree' - Disagree: otherwise
Q48	General vaccine	Getting vaccinated against COVID-19 is an excellent way to protect myself from this disease.	- Strongly Disagree, - Disagree, - Undecided/Neutral, - Agree, - Strongly Agree	- Agree: 'Agree' and 'Strongly Agree' - Disagree: otherwise
Q54	COVID-19 vaccine	If a COVID19 vaccine becomes available for me, I will get it.	- Strongly Disagree, - Disagree, - Undecided/Neutral, - Agree, - Strongly Agree	- Agree: 'Agree' and 'Strongly Agree' - Disagree: otherwise

residential units within each selected sector, we determined the appropriate number of villages using a probability method. Subsequently, households within each selected village were surveyed until the desired sample size was attained. Given the challenges posed by the COVID-19 pandemic, which restricted random sampling methods, we employed a convenient sampling technique. This adaptation was necessary to ensure the feasibility of reaching our desired sample size under the prevailing circumstances.

Study setting

In early January 2022, we administered pilot-tested structured questionnaires to 10 adult caregivers. After our pilot test, we embarked on conducting a cross-sectional study from January to March 2022 to assess COVID-19 vaccine acceptance with respondents' characteristics. Data from these respondents was used to assess internal consistency reliability using Cronbach's alpha (α) [31–33]. The results showed adequate internal consistency reliability (with Cronbach's $\alpha=0.72$) [32, 33].

Respondent recruitment

The study shows targeted Rwandans who were mothers or caregivers to children in the vaccination bracket. Respondents were recruited using snowball and convenient sampling techniques, resulting in a sample size of 2,126 respondents that included both men and women, with a 100% response rate. Each respondent underwent

an interview, facilitated by trained research assistants to accommodate those with low literacy levels. To ensure the integrity of data collection, research assistants were rigorously selected based on their knowledge of research ethics and data collection techniques, each was assigned a unique code and assessed prior engagement. Before the sampling, they received comprehensive training on study objectives, probing techniques, and data recording protocols. Additionally, five data quality control officers, one per district, were also trained in questionnaire data types and Microsoft Excel usage.

Respondents were recruited through face-to-face interviews conducted in 50 randomly selected villages (Fig. 1), with confidentiality guaranteed throughout the process. The NGOs played a significant role in capacity building, education, and combating emerging infectious diseases through immunisation, from the genocide through the Ebola virus outbreak to the COVID-19 pandemic. Collaboration between the NGOs and local administration facilitated an efficient data collection process.

Dependent and independent variables

Our independent variables included age, sex, marital status, religion, level of education, occupation, and monthly income. Our Outcome variables included COVID-19 vaccine acceptance, acceptance of other vaccines, and overall vaccine acceptance by mothers or caregivers in Rwanda. All these vaccine acceptances were assessed

based on responses to five questions of the questionnaire (Questions 30, 36, 44, 48 and 54; Supplementary file 1).

The potential responses were graded on a five-point Likert scale [34], and recorded as 'Strongly Disagree', 'Disagree', 'Undecided/Neutral', 'Agree' and 'Strongly Agree'. All the responses were then processed and rated on a two-point rating scale as 'Agree' and 'Disagree'; wherein the 'Neutral' category was considered as 'Disagree' (Table 1).

COVID-19 vaccine acceptance was evaluated with the use of one question (Q54; Supplementary file 1), while acceptance of vaccines other than COVID-19 vaccines was evaluated with the use of four questions (Q30, Q36, Q44, and Q48; Supplementary file 1). The overall VA was evaluated by combining the C19VA and the acceptance of the other vaccines in general.

Data collection

The questionnaire was developed using standard tools and methods obtained from WHO which was administered by trained research assistants, with all responses from the respondents recorded on paper. The questionnaire items were derived from WHO vaccination coverage cluster surveys: reference manual (version updated July 2015), which guided our data collection process. Demographic information collected included age, sex, occupation, and other pertinent details.

Data entry was managed using a Microsoft Excel spreadsheet designed specifically for this purpose. Trained research assistants conducted structured, face-to-face interviews with mothers/caregivers using a pre-piloted questionnaire based on the 2015 WHO standard version of vaccination coverage cluster surveys. Following data collection, field supervisors reviewed all the data from various districts to ensure its accuracy and completeness [29].

Statistical analysis

The analysis of data was performed using CDC Epi Info (Centre for Disease Control, Georgia, USA) version 7.2.5.0. Categorical variables were summarised as proportions, while continuous variables were presented as means with standard deviations. Associations between respondents' characteristics and outcome variables were evaluated using the Pearson Chi-square (χ^2) test.

To further establish associations or differences between vaccine acceptance and respondents' general characteristics, multivariate logistic regression was employed. Additionally, to mitigate potential confounding effects from unobserved district-level variables on vaccine acceptance, we also conducted the adjusted logistic regression by including Districts' fixed effects as a robustness check. The significant level threshold on both sides was set at 0.05.

Results

Statistical summary of the overall population's demographic assessment

Table 2 presents a descriptive summary of the demographic characteristics of the study population, which comprised 2,126 respondents. A majority (82.2%, 95% C.I; 80.5–83.8) of respondents were females, about half (51.4%, 95% C.I; 49.3–53.5) had completed their primary education, and slightly more than three-quarters (78.7%, 95% C.I; 76.9–80.4) were married.

Demographic summaries

Nearly half of the respondents were 30 years old or younger (49.4%, 95% CI: 43.7–51.5). In terms of relationship with the child(ren), more than four-fifths [1,805 (84.9%, 95% C.I; 83.3–86.4)] of the respondents were mothers of the child(ren), only 321 (15.1%, 95% C.I; 13.6–16.7) of the respondents were caregivers. As for the districts, Nyagatare had the highest proportion of respondents, at 23.2% (494/2,126), and Nyamagabe had the lowest percentage, at 14.7% (313/2,126). Four hundred and seventy-three (22.3%) of the respondents were sampled from Nyarugenge, 456 (21.5%) from Ngoma and 390 (18.3%) from Ngororero.

The mean age of the study's respondents was 31 years old (range: 18–58). All the respondents' characteristics were significantly ($p < 0.05$) associated with the districts.

Acceptance of adults vaccination assessment

Our analysis revealed a COVID-19 acceptance of 1,948 out of 2,126 respondents who participated in the study. On a binary scale, respondents had varying opinions on the acceptance of vaccines; with 91.6–98.2% range of vaccine acceptance rate and 1.8–8.4% vaccine hesitance rate (Supplementary file 2). Two thousand and eighty-seven (98.2%, 95% C.I; 97.5–98.7), asserted that vaccines are of importance to their health, 2,049 (96.4%, 95% C.I; 95.5–97.1) agreed that getting vaccinated protects them from diseases, while 2,026 (95.3%, 95% C.I; 94.3–96.1) agreed that getting vaccinated with the COVID-19 vaccines is important for the health of others in the community (Supplementary file 1).

Table 3 presents the association between respondents' characteristics with COVID-19 and vaccine acceptance for other illnesses. It also shows a COVID-19 vaccine acceptance was 1,948, while the general vaccine acceptance was 2,101.

Our results revealed that respondents aged 31–40 years, caregivers, and married individuals were more likely to accept COVID-19 vaccines. Specifically, those aged 31–40 years were 1.2 times more likely (OR; 1.2, 95% C.I; 0.8–1.6), caregivers were 2.9 times more likely (OR; 2.9, 95% C.I; 1.4–5.9), and married individuals were 1.3 times more likely (OR; 1.3, 95% C.I; 0.9–2.1) to accept

Table 2 Demographic summaries of the study population with district-wise characteristics of the respondents ($N=2,126$)

Characteristic	District							χ^2	p-value
	Total	Percent (95% C.I)	Ngoma	Ngororero	Nyagatare	Nyamagabe	Nyarugenge		
Age groups (in years)									
≤ 30	1,050	49.4 (47.3–51.5)	223	165	279	157	226	46.738	< 0.001*
31–40	920	43.3 (41.2–45.4)	182	187	198	147	206		
> 41	156	7.3 (6.3–8.5)	51	38	17	9	41		
Mean age ($\bar{x} \pm$ pm SD)	31.0 ± 6.6		31.0 ± 7.1	32.1 ± 6.0	29.0 ± 5.0	30.0 ± 5.0	31.0 ± 7.1		
Sex									
Male	378	17.8 (16.2–19.5)	100	126	78	6	68	120.643	< 0.001*
Female	1,748	82.2 (80.5–83.8)	356	264	416	307	405		
Relation with child(ren)									
Mother	1,805	84.9 (83.3–86.4)	390	279	417	293	426	82.908	< 0.001*
Caregiver	321	15.1 (13.6–16.7)	66	111	77	20	47		
Religion									
Catholic	81	3.8 (3.1–4.7)	8	3	12	25	33	45.513	< 0.001*
Protestant	2,045	96.2 (95.3–96.9)	448	387	482	288	440		
Marital status									
Not married	452	21.3 (19.6–23.1)	95	58	104	86	109	17.694	1.4 × 10 ^{−3} *
Married	1,674	78.7 (76.9–80.4)	361	332	390	227	364		
Educational status									
NFE	241	11.3 (10.1–12.7)	70	23	58	65	25	397.999	< 0.001*
Primary	1,093	51.4 (49.3–53.5)	322	268	265	111	127		
Secondary	719	33.8 (31.8–35.9)	59	95	158	126	281		
Tertiary	73	3.4 (2.7–4.3)	5	4	13	11	40		
Occupation									
Artisan	1,351	63.5 (61.5–65.6)	434	303	282	278	54	981.360	< 0.001*
Casual labour	223	10.5 (8.6–11.2)	6	40	46	28	103		
Civil Servant	111	5.2 (5.0–6.9)	3	22	50	1	35		
Unemployed	441	20.7 (19.1–22.5)	13	25	116	6	281		
Total (%)	2,126	100.0	456 (21.5)	390 (18.3)	494 (23.2)	313 (14.7)	473 (22.3)		
National population	1,814,030 (100.0)		404,048 (22.3)	367,955 (20.3)	653,861 (36.0)	371,501 (20.5)	16,665 (0.9)		

*Significant p value was ≤ 0.05 , %; proportion of respondents, 95% C.I.; 95% Confidence interval, NFE; No Formal Education, SD; Standard Deviation

COVID-19 vaccines than their respective counterparts (Table 3).

Regarding general vaccine acceptance, significant associations were observed with respondents' sex and marital status only after adjusting for district differences.

Logistic regression analysis, adjusted for the district of respondents, indicated that respondents in the 3–40 years vs. the ≥ 41 years groups and those that were married were, respectively, one and a fifth (adjusted OR; 1.2, 95% C.I.; 0.7–1.9) vs. one and a half (aOR; 1.5, 95% C.I.; 0.6–3.6), and three and a half (aOR; 3.5, 95% C.I.; 1.5–8.2), more likely to accept all vaccines when compared with respondents who were ≤ 30 years old, and those who were not married (Table 3).

This study revealed that the proportion of COVID-19 vaccine acceptance was 1,948 (91.6%, 95% C.I.; 90.4–92.7),

acceptance of other vaccines of 2,101 (98.2%, 95% C.I.; 97.5–98.7), and the acceptance of all vaccines of 1,935 (91.0%, 95% C.I.; 89.7–92.2). Thus, there was a COVID-19 of 178 (8.4%, 95% C.I.; 7.3–9.6), and an overall vaccine hesitancy of 191 (8.9%, 95% C.I.; 7.8–10.3).

Bivariate analyses revealed that sex, relationship with child(ren), religion, education, occupation, and District were significantly ($p < 0.05$) associated with COVID-19. Only the respondents' District was significantly associated with other vaccine acceptance, while sex, relationship with the child(ren), religion, education, occupation, and District were significantly ($p < 0.05$) associated with the acceptance of all vaccines (Supplementary file 1).

Table 3 Association of respondents' characteristics with COVID-19 vaccine acceptance and general vaccine acceptance

S/N	Characteristic	COVID-19 vaccine acceptance (n = 1,948)				Vaccine acceptance (n = 2,101)			
		n(%)	OR (95% C.I.)	aOR (95% C.I.)	p-value	n(%)	OR (95% C.I.)	aOR (95% C.I.)	p-value
1.	Age (in years)								
	≤ 30 (Ref)	963 (49.4)	1.0	1.0	-	1,039 (49.5)	1.0	1.0	-
	31–40	836 (42.9)	1.2 (0.8–1.6)	1.2 (0.9–1.4)	1.0×10^{-1}	908 (43.2)	1.2 (0.5–2.6)	1.2 (0.7–1.9)	5.2×10^{-1}
	≥ 41	149 (7.6)	0.6 (0.3–1.5)	0.4 (0.3–0.7)	2.7×10^{-3}	154 (7.3)	1.3 (0.3–6.6)	1.5 (0.6–3.6)	3.6×10^{-1}
2.	Sex								
	Female (Ref)	1,574 (80.8)	1.0	1.0	-	1,724 (82.1)	1.0	1.0	-
	Male	374 (19.2)	0.05 (0.02–0.2)	0.02 (0.01–0.06)	< 0.001	377 (17.9)	0.2 (0.02–1.2)	0.2 (0.06–0.6)	6.5×10^{-3}
3.	Relationship								
	Mother (Ref)	1,642 (84.3)	1.0	1.0	-				
	Caregiver	306 (15.7)	2.9 (1.4–5.9)	3.8 (2.6–5.5)	< 0.001	-	-	-	-
4.	Religion								
	Catholic (Ref)	64 (3.3)	1.0	1.0	-				
	Protestant	1,884 (96.7)	0.4 (0.2–0.7)	0.4 (0.3–0.6)	< 0.001	-	-	-	-
5.	Marital status								
	Not married (Ref)	421 (21.6)	1.0	1.0	-	450 (21.4)			
	Married	1,527 (78.4)	1.3 (0.9–2.1)	1.4 (1.1–1.7)	8.8×10^{-3}	1,651 (78.6)	3.4 (0.8–14.5)	3.5 (1.5–8.2)	3.4×10^{-3}
6.	Educational status								
	Tertiary (Ref)	66 (3.4)	1.0	1.0	-				
	Secondary	614 (31.5)	0.9 (0.3–2.3)	0.7 (0.4–1.2)	2.6×10^{-1}	-	-	-	-
	Primary	1,045 (53.6)	0.2 (0.1–0.5)	0.1 (0.08–0.2)	< 0.001	-	-	-	-
	NFE	223 (11.5)	0.3 (0.1–0.9)	0.3 (0.1–0.5)	< 0.001	-	-	-	-
7.	Occupation								
	Artisanal (Ref)	1,213 (62.3)	1.0	1.0	-				
	Causal Labour	206 (10.6)	0.4 (0.2–0.7)	0.3 (0.2–0.4)	< 0.001	-	-	-	-
	Civil Servant	107 (5.5)	0.1 (0.04–0.4)	0.06 (0.03–0.1)	< 0.001	-	-	-	-
	Unemployed	422 (21.7)	0.2 (0.1–0.4)	0.1 (0.08–0.2)	< 0.001	-	-	-	-

OR; Odds ratio, aOR; adjusted Odds ratio, 95% C.I.; Confidence Interval; **Boldface** numbers indicate significant *p*-values, Ref; Reference

NB: For the vaccine acceptance (VA) with *n*=2,101 respondents, the relationship, educational status as well as occupation of the respondents could not fit well into the logistic regression model and thus, were not included in the model

Discussion

Our findings revealed that the importance of COVID-19 vaccination for personal health and community well-being was recognized by the majority of respondents. Having a relationship with the child(ren) was the only characteristic significantly associated with COVID-19 vaccine acceptance (*p*; 3.2×10^{-3} , OR; 2.9, 95% C.I; 1.4–5.9).

We investigated the acceptance and, consequently, the hesitancy of COVID-19 vaccines as well as other vaccines during the COVID-19 pandemic and identified factors related to this acceptance and hesitancy using a face-to-face survey of four out of five different provinces in Rwanda. The results indicated that COVID-19 vaccine acceptance varied among different demographic characteristics within the study population. Our study also reported COVID-19 vaccine acceptance and vaccine hesitancy in low proportions amongst the respondents. The adjusted models did not attenuate all the odds ratios for COVID-19 vaccine covariates, which is not surprising due to the optimal district sample sizes and strong correlations among the covariate items. The high correlations

among the COVID-19 vaccine survey items suggest motivated reasoning as well as the effective efforts of the MoH and stakeholders. Summary of findings: Our findings indicated that COVID-19 Vaccine acceptance was 91.6% (95% C.I; 90.4–92.7), with a COVID-19 vaccine hesitancy rate of 8.4%. This high vaccination acceptance aligns with previously reported rates in China (91.3%) and among healthcare workers in Germany (91.7%) [35, 36]. However, our finding was relatively lower than the reported COVID-19 vaccine acceptance rates in Niger, Indonesia, and Malaysia (93.3%) [37–39], and Ecuador (97.0%) [36]. In comparison, our finding was higher than the 82.6% reported among community-dwelling populations in China [40], the 80.3% reported in a global study of ten low- to middle-income countries [17], the 79.1% rate reported among 23 countries in Asia, Africa, Europe and America [41], the universal coverage rate of 70% [42], the 65% rates in the USA [43], 65.2% in Nigeria [44], and significantly higher than the 25.7–37.4% reported among Bosnia and Herzegovina, Cameroon and Jordanians [45, 46]. The high COVID-19 vaccine acceptance rate in The high COVID-19 vaccine rate reported

in Rwanda is attributed to increased vaccine awareness campaigns conducted by Rwanda's Ministry of Health across districts. These factors influenced the positive outcome of our study which aligns with the study objectives. Differences observed among various studies could also be attributed to other factors, variations in study designs and approaches to tackling COVID-19 vaccine acceptance and hesitancy. Rwanda's comprehensive mass vaccination and education campaigns have likely contributed significantly to reaching and educating the entire population.

When comparing educational levels, the willingness to receive the COVID-19 vaccine was lower among those with no formal education or only primary education compared to those with secondary or tertiary education. Specifically, individuals with no formal education were significantly less likely to accept the COVID-19 vaccine (OR; 0.2, 95% C.I.; 0.1–0.5 vs. OR; 0.3, 95% C.I.; 0.1–0.9). Our study findings align with the results of another study conducted in western Ethiopia [47].

Numerous studies have demonstrated that parents who are hesitant about vaccinations are generally well-informed, concerned with health issues, and proactively seeking health advice [48, 49]. Higher levels and lower levels of educational attainment and socioeconomic status have been linked to higher vaccine hesitancy, highlighting the complex interplay of various factors influencing vaccine acceptance and hesitancy [50].

Significance of COVID-19 vaccines to respondents

In our study, 95.8% of respondents understood the importance of vaccination and agreed that getting vaccinated against COVID-19 was an excellent way to protect themselves. Our analysis revealed that middle-aged respondents (31–40 years), caregivers, and married individuals were more likely to accept COVID-19 vaccines than their counterparts. These findings were consistent with many other studies reported worldwide [51–53], but differ from a study conducted in China, where females, older individuals, and those with lower educational attainment had more positive attitudes towards COVID-19 vaccines [40]. Similarly, our findings contrast with a study in Jordan, which reported that older generations and females were less likely to accept the COVID-19 vaccines [45].

Deep concern about the risk of COVID-19 infection are driving factor for front-line health workers and the public to accept COVID-19 vaccines in Rwanda. COVID-19 vaccine hesitancy in Rwanda is due to misinformation on the SARS-CoV-2 virus that can affect people in many ways and has life-threatening consequences, especially for the older generation and can also severely affect young and healthy people [54, 55].

According to the World Bank Development Economics Data Group, as of July 2021, approval of the free COVID-19 vaccine was substantial in six sub-Saharan African countries, with four out of five (80%) people eager to receive the inoculation [56]. This was lower than our finding of 91.6% COVID-19 vaccine acceptance and 98.2% general vaccine acceptance rates. Some general characteristics and communities/districts that influence vaccine acceptance in other studies are synonymous with those reported elsewhere, such as in Rwanda, Cameroon, and Jordan [37, 45, 57]. Political, social, and psychological considerations play a role in individual vaccine decisions [58].

Our findings revealed a 98.2% acceptance rate for various other vaccines, which was comparable to the 98% acceptance rate for the first dose of the Human papillomavirus (HPV) vaccine among targeted school-age girls between 2011 and 2018 across Rwandan provinces [59]., and higher than the 90% acceptance rate reported among children vaccinated against polio in Rwanda [60].

Comparison of COVID-19 vaccine acceptance with other studies

Our vaccine acceptance rate was high compared to the 44.3% reported for the hepatitis B vaccine among students of the Makerere University of Uganda [61], and very high compared to the adherence and 24-month completion rates of 27.0–44.8% for the various doses of hepatitis A and B in the USA [62]. In our study, vaccine acceptance was associated with sex, which is in line with that of Makerere University in Uganda [61]. There were however differences, arising from the different study designs; our study had a larger sampling space while the Uganda study was restricted to University students.

Our vaccine hesitancy rate of 1.8%, is comparable to the 1.5% reported among unvaccinated children in Canada [63]. Another study in Rwanda identified service unavailability and inadequate follow-up as primary factors influencing vaccine hesitancy [37], aspects not explored in our study. The slight differences between COVID-19 vaccine acceptance and vaccine hesitancy in general in our study as well as other studies, are justified by distrust as well as complacency, convenience, and confidence [11].

In-depth analysis of factors contributing to vaccine acceptance and vaccine hesitancy.

The WHO has acknowledged the increasing crisis of vaccine hesitancy affecting countries with varying resource levels [64]. Concerns regarding vaccine safety, perceived vaccine effectiveness, and needle-point anxiety significantly contribute to vaccine hesitancy [65]. This complex phenomenon is influenced by contextual factors including time, place, and specific vaccines [66]., largely stemming from insufficient knowledge about vaccines [67, 68]. Effective strategies to address vaccine

concerns should leverage digital platforms, shifting from defensive approaches to proactive public health campaigns that emphasise the importance of vaccines [67, 68]. Given the influence of social media on vaccine hesitancy, there is a critical need to develop digital communication techniques to counter misinformation propagated by anti-vaccine groups. Further, healthcare professionals play a crucial role in shaping vaccine acceptance [69]; therefore, they should be well-informed and skilled in addressing COVID-19-related inquiries to mitigate immunisation challenges [70]. This study represents the first assessment of the COVID-19 vaccine acceptance among mothers or caregivers in Rwanda, underscoring a significantly high proportion of respondents accepting the vaccine. The lessons learnt from this study can be emulated by other countries in the fight against the COVID-19 pandemic. Further research may be conducted based on country-specific context.

This study could not capture the changes or trends in COVID-19 vaccine acceptance and hesitancy over time or in response to specific events or interventions. Future research to prioritize other COVID-19 vaccine intervention programs in Rwanda employing longitudinal or experimental study designs would be necessary to examine the causal relationships and temporal dynamics in COVID-19 vaccine acceptance and hesitancy.

Strengths and limitations of the study

First, unlike most previous studies that relied on online surveys or convenience samples, this study used a community-based face-to-face survey with a multi-stage random sampling technique to collect data from five districts in Rwanda. We used random sampling when we wanted to find the number of desired provinces, districts sectors and villages to include in our study areas. This method ensured a high response rate and a representative sample of the population. Second, the data used in this study was collected by trained field workers who had mastery of the sectors, cells, and villages in the study areas. The district mayors were consulted before the commencement of the study for approval to conduct our research in their administrative areas. These steps enhanced the quality and credibility of the data collection process. Third, the study used a comprehensive questionnaire that measured various aspects of COVID-19 vaccine acceptance and hesitancy, such as knowledge, attitude, belief, trust, and intention. The questionnaire incorporated the validated 5 C scale, providing a robust framework to assess the psychological factors influencing vaccine hesitancy [9]. The use of this questionnaire enabled the study to capture the complexity and diversity of the factors affecting COVID-19 vaccine acceptance and hesitancy in Rwanda.

However, this study also has some limitations. First, this study utilized a cross-sectional design, assessing

the COVID-19 vaccine acceptance and hesitancy at a single point in time. In this regard, we used a convenient sampling technique to obtain the required sample size because this study was conducted during the COVID-19 pandemic, where we anticipated challenges and we found some homes closed and failed to access information.

As such, it could not capture the changes or trends in COVID-19 vaccine acceptance and hesitancy over time or in response to specific events or interventions. Future research employing longitudinal or experimental study designs would be necessary to examine the causal relationships and temporal dynamics in COVID-19 vaccine acceptance and hesitancy. Second, this study was conducted in five randomly selected districts of Rwanda, potentially limiting the generalizability of its findings to other districts or regions within the country. Variations in cultural, social, economic, and political differences between Rwanda and other countries further restrict the applicability of the results. Therefore, caution should be exercised when extrapolating or comparing these study results to different contexts or populations. Third, this study relied on self-reported data from the respondents, which may introduce recall bias, social desirability bias, or measurement error. Respondents may struggle to accurately recall or report their actual knowledge, attitude, belief, trust, or intentions regarding the COVID-19 vaccine, potentially aligning responses with perceived social norms or researcher/authorities expectations. Moreover, the questionnaire's validity, reliability, and sensitivity of the items or the scales in measuring COVID-19 vaccine acceptance and hesitancy could be subject to limitations. Future research could benefit from incorporating more objective or verifiable data sources such as vaccination records. Moreover, employing refined and standardized instruments would enhance the comparability and consistency of findings across studies. This study had a weakness in identifying all papers on COVID-19 vaccine acceptance and vaccine hesitancy from region to region due to country-specific policy measures towards COVID-19 vaccination. Therefore, factors influencing vaccine acceptance and hesitancy on COVID-19 did not affect countries in the same way due to the nature of individual and state-level settings.

Conclusion

We investigated the acceptance and, consequently, the hesitancy of COVID-19 vaccines and other vaccines during the COVID-19 pandemic and identified factors related to this acceptance and hesitancy using a face-to-face survey of four out of five different provinces in Rwanda. This study found a remarkably high rate of COVID-19 vaccine acceptance among mothers and caregivers in Rwanda, with an overall acceptance rate of 91.6% and a general vaccine acceptance (VA) rate of

98.2%.“These findings underscore the importance of tailored public health interventions and provide a model for other nations to enhance vaccine uptake and combat hesitancy in the fight against COVID-19. Notably, having a relationship with the child(ren) was the only characteristic significantly associated with COVID-19 vaccine acceptance (p ; 3.2×10^{-3} , OR; 2.9, 95% C.I; 1.4–5.9). To our knowledge, this study represents the first assessment of the COVID-19 vaccine acceptance among mothers or caregivers in Rwanda, Rwanda’s proactive approach to infectious disease preparedness offers valuable lessons for other countries facing similar health and political challenges.

The findings of this study in Rwanda in a broader perspective will guide other countries that may have some sort of implications in addressing COVID-19 vaccine acceptance and hesitancy based on the health policy of specific countries. Further research may be conducted based on country-specific context.

Abbreviations

95% C.I	95% Confidence Interval
COVID – 19	Coronavirus Disease 2019
C19VA	COVID – 19 Vaccine Acceptance
C19VH	COVID-19 Vaccine Hesitancy
MoH	Ministry of Health
NFE	No Formal Education
OR	Odds Ratio
SARS-CoV-2	Severe Acute Respiratory Syndrome Coronavirus
VA	Vaccine Acceptance
VH	Vaccine Hesitancy
WHO	World Health Organisation
χ^2	Pearson’s Chi-square

Supplementary Information

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Supplementary Material 1

Supplementary Material 2

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Author contributions

Conceptualization: E M, C M D, S A M, S C. Data curation: E M, F N C. Formal analysis: E M, F N C. Investigation: E M. Methodology: E M, F Y, F N C, P C K E, S C. Project administration: E M, C M D, S C. Resources: E M, M D, S C. Supervision: S A M, C M D, S C. Validation: E M, F Y, M D C, F N C, Q C, S A M, C M D, P C K E, W M, R M, I E M, S C. Writing – original draft: E M, F N C, S A M, C M D, M W, S C. Writing – review and editing: E M, F Y, M C D R, F N C, S A M, C M D, P C K E, W M, R M, I E M, S C.

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Data availability

The field data supporting this study’s findings are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

This study was conducted in strict conformity with the Helsinki Declaration and cleared by the Institutional Review Board (IRB) of the University of Rwanda, College of Medicine and Health Science (No. 402/CMHS IRB/2020) and the Ethics Committee of the University of Heidelberg ethical committee (S-829/2021). The Executives of each sector were approached with the objectives of the study, to obtain verbal administrative clearance. Informed consent has been obtained from the respondents, their parents and legally authorized representatives in this study.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Disclosure

I declare that this study has not been done by anyone other than the primary author.

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